

Scenario 1

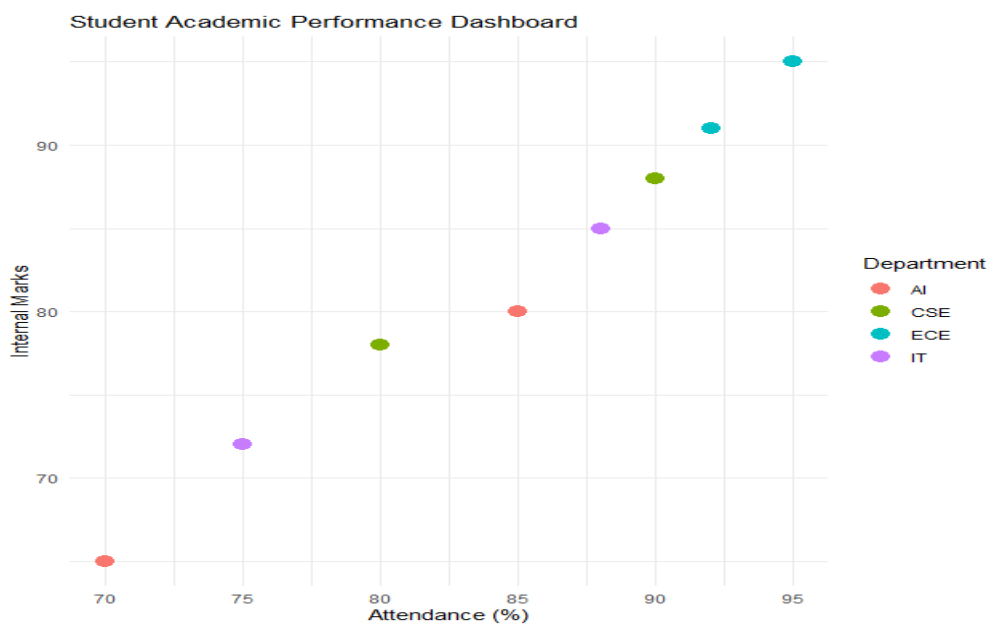
Title

Student Academic Performance Dashboard

CODE:

```
library(ggplot2)
student <- data.frame(
  Attendance = c(90, 85, 75, 95, 80, 70, 88, 92),
  Marks = c(88, 80, 72, 95, 78, 65, 85, 91),
  Department = c("CSE", "AI", "IT", "ECE", "CSE", "AI", "IT", "ECE")
)
p <- ggplot(student, aes(x = Attendance, y = Marks, color = Department)) +
  geom_point(size = 4) +
  labs(
    title = "Student Academic Performance Dashboard",
    x = "Attendance (%)",
    y = "Internal Marks"
  ) +
  theme_minimal()
print(p)
ggsave(
  filename = "Scenario1_Output.png",
  plot = p,
  width = 6,
  height = 4,
  dpi = 300
)
```

OUTPUT:



Scenario 2

Title

Electric Vehicle Market Analysis

CODE :

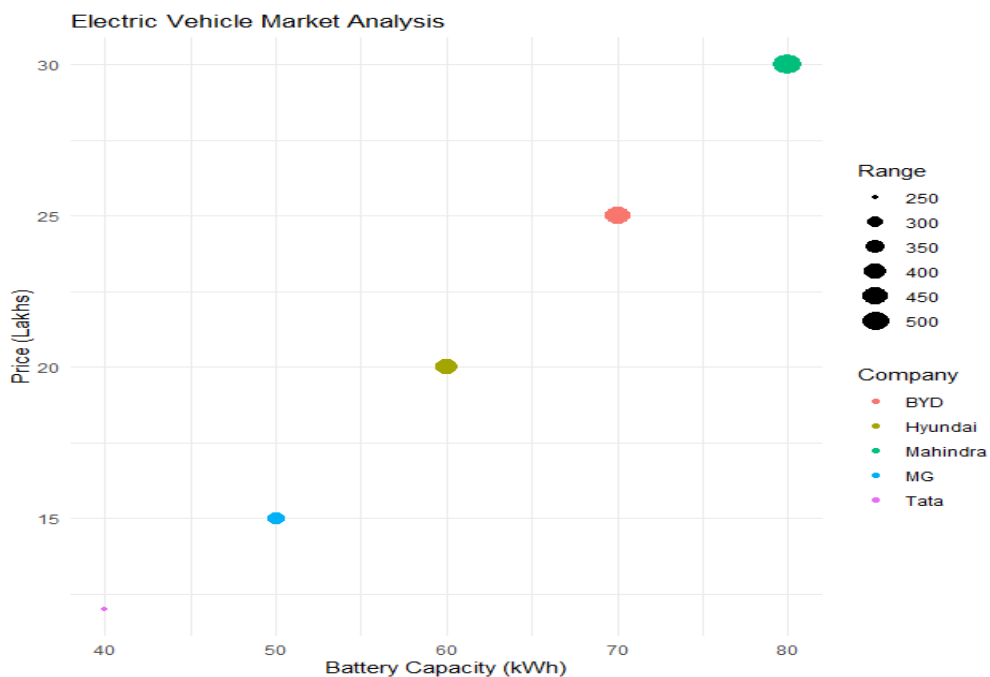
```
library(ggplot2)

ev <- data.frame(
  Battery = c(40,50,60,70,80),
  Price = c(12,15,20,25,30),
  Range = c(250,320,380,450,520),
  Company = c("Tata","MG","Hyundai","BYD","Mahindra")
)

p <- ggplot(ev, aes(Battery, Price, size=Range, color=Company)) +
  geom_point() +
  labs(title="Electric Vehicle Market Analysis",
       x="Battery Capacity (kWh)",
       y="Price (Lakhs)") +
  theme_minimal()

print(p)
ggsave("Scenario2_Output.png", p)
```

OUTPUT:



Scenario 3

Title

Hospital Patient Admission Analysis

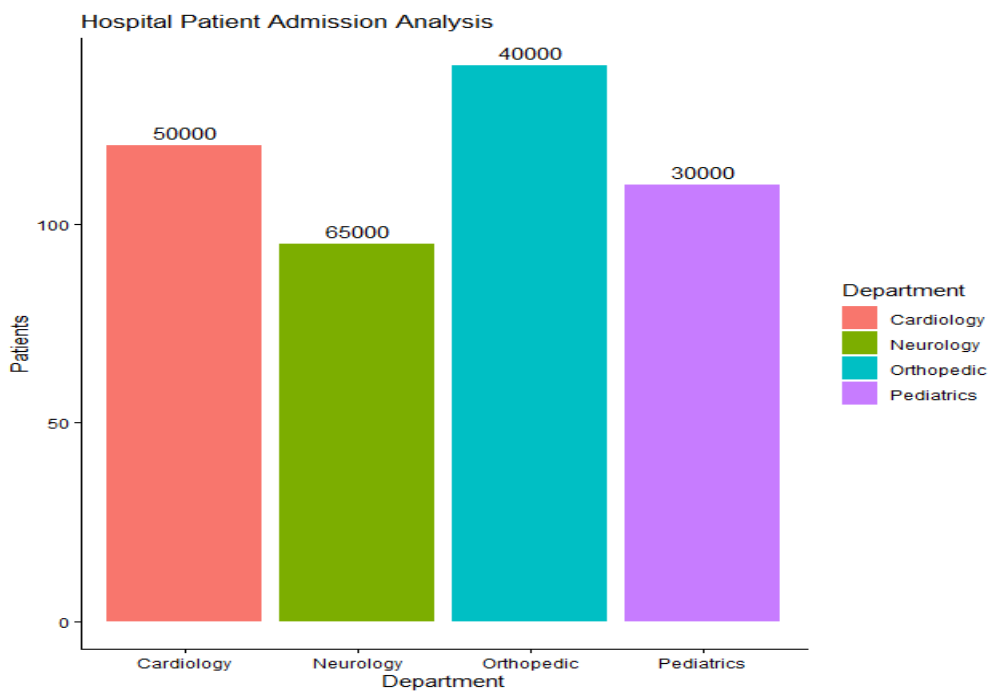
Code:

```
library(ggplot2)

hospital<-data.frame(
  Department=c("Cardiology","Neurology","Orthopedic","Pediatrics"),
  Patients=c(120,95,140,110),
  Cost=c(50000,65000,40000,30000)
)

ggplot(hospital,aes(Department,Patients,fill=Department))+
  geom_col()+
  geom_text(aes(label=Cost),vjust=-0.5)+
  labs(title="Hospital Patient Admission Analysis")+
  theme_classic()
```

OUTPUT:



Scenario 4

Title

Weather Monitoring System

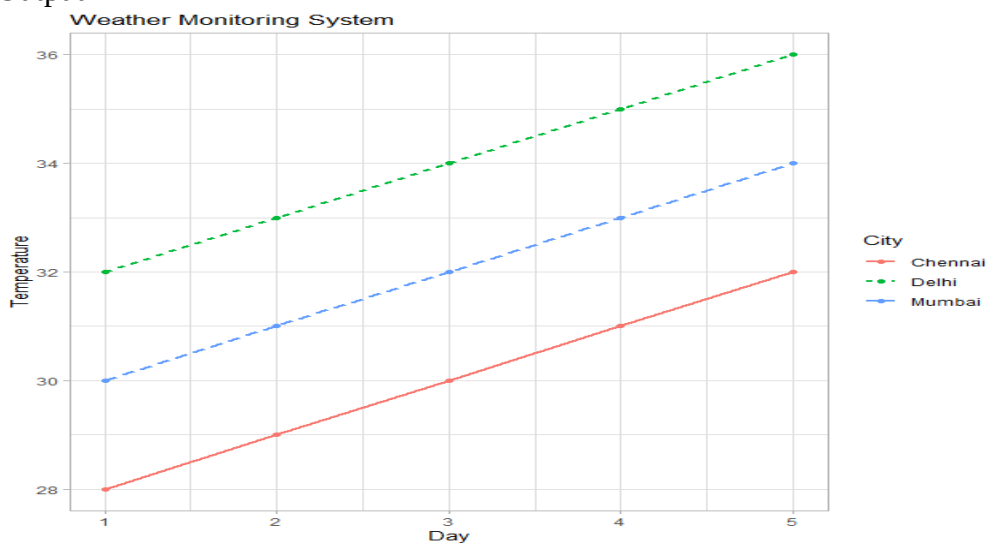
Code:

```
library(ggplot2)

weather<-data.frame(
  Day=rep(1:5,3),
  Temperature=c(32,33,34,35,36,30,31,32,33,34,28,29,30,31,32),
  City=rep(c("Delhi","Mumbai","Chennai"),each=5)
)

ggplot(weather,aes(Day,Temperature,color=City,linetype=City))+
  geom_line(linewidth=1)+
  geom_point()+
  labs(title="Weather Monitoring System")+
  theme_light()
```

Output



Scenario 5

Title

Supermarket Sales Analysis

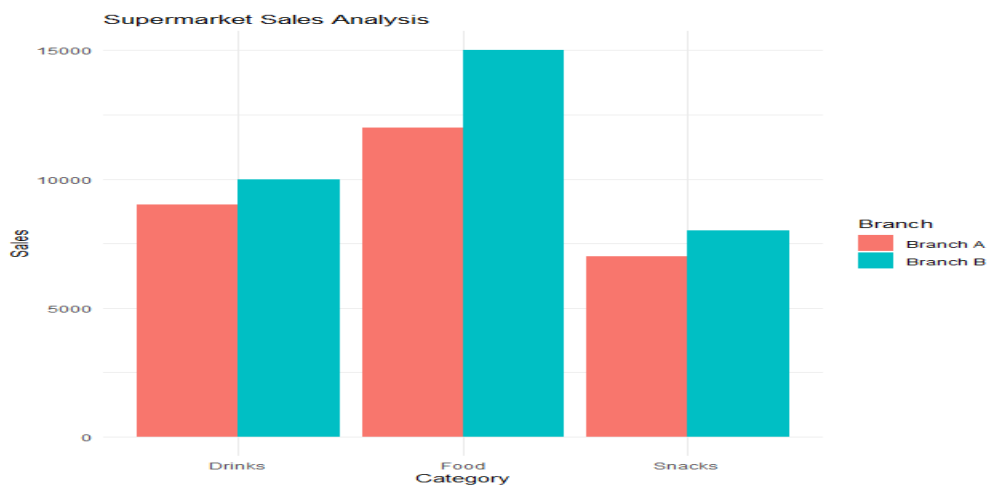
Code:

```
library(ggplot2)

sales<-data.frame(
  Category=rep(c("Food", "Drinks", "Snacks"),2),
  Branch=rep(c("Branch A", "Branch B"),each=3),
  Sales=c(12000,9000,7000,15000,10000,8000)
)

ggplot(sales,aes(Category,Sales,fill=Branch))+
  geom_col(position="dodge")+
  labs(title="Supermarket Sales Analysis")+
  theme_minimal()
```

OUTPUT:



Scenario 6

Title

Employee Salary Distribution

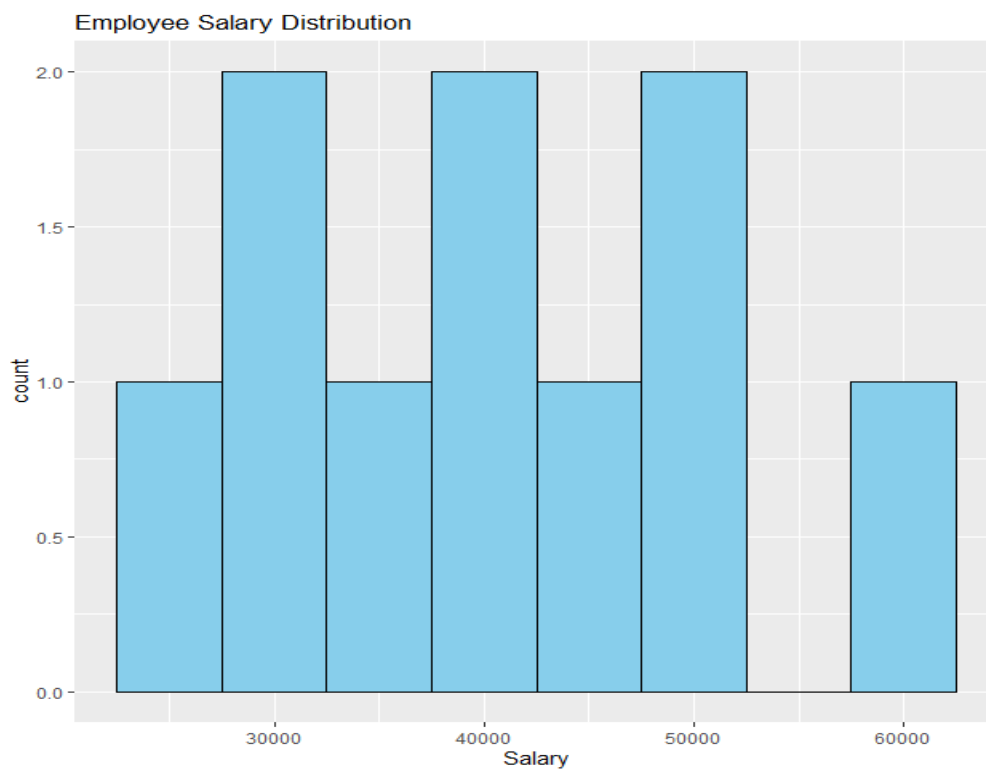
CODE:

```
library(ggplot2)

salary<-data.frame(
Salary=c(25000,28000,30000,35000,40000,42000,45000,48000,50000,60000)
)

ggplot(salary,aes(Salary))+
geom_histogram(binwidth=5000,fill="skyblue",color="black")+
labs(title="Employee Salary Distribution")+
theme_gray()
```

OUTPUT:



Scenario 7

Title

Smartphone Market Comparison

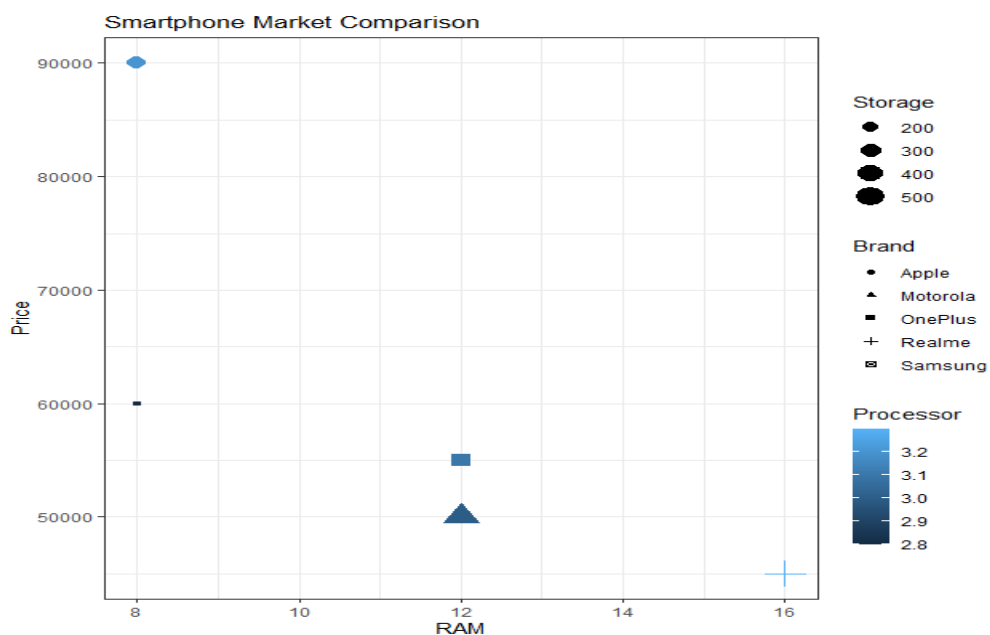
CODE:

```
library(ggplot2)

phone<-data.frame(
  Brand=c("Samsung","Apple","OnePlus","Realme","Motorola"),
  RAM=c(8,8,12,16,12),
  Price=c(60000,90000,55000,45000,50000),
  Storage=c(128,256,256,512,512),
  Processor=c(2.8,3.2,3.1,3.3,3.0)
)

ggplot(phone,
  aes(RAM,Price,
  shape=Brand,
  size=Storage,
  color=Processor))+
  geom_point()+
  labs(title="Smartphone Market Comparison")+
  theme_bw()
```

OUTPUT:



Scenario 8

Title

Online Movie Rating Analysis

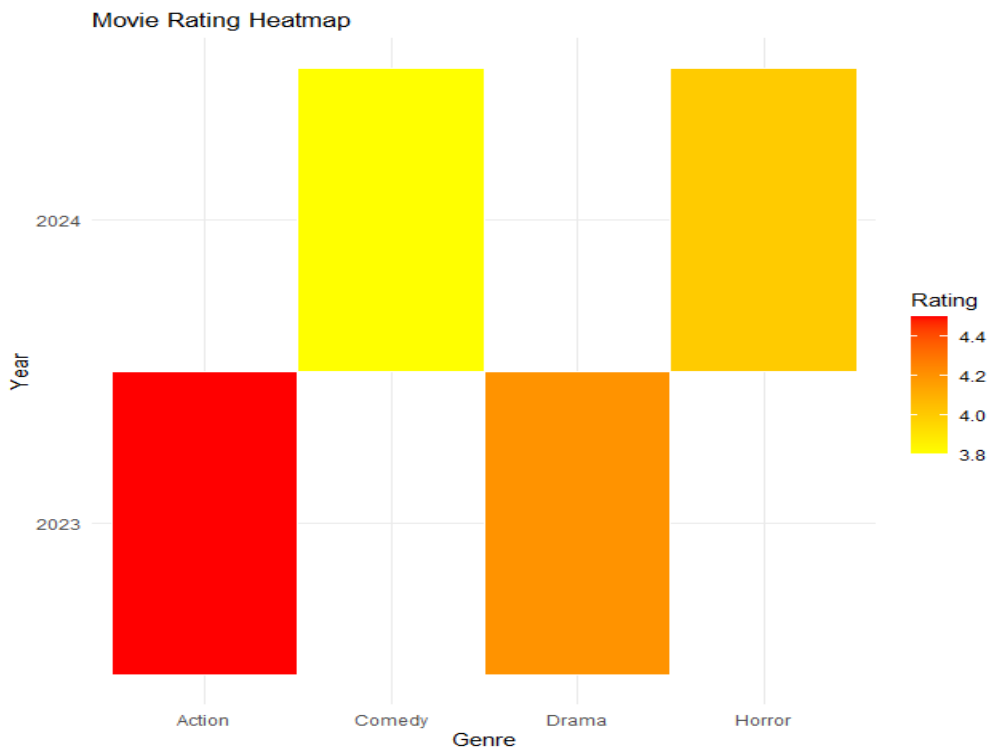
CODE:

```
library(ggplot2)

movie<-data.frame(
  Genre=c("Action","Drama","Comedy","Horror"),
  Year=c("2023","2023","2024","2024"),
  Rating=c(4.5,4.2,3.8,4.0)
)

ggplot(movie,aes(Genre,Year,fill=Rating))+
  geom_tile(color="white")+
  scale_fill_gradient(low="yellow",high="red")+
  labs(title="Movie Rating Heatmap")+
  theme_minimal()
```

OUTPUT:



Scenario 9

Title

Code:

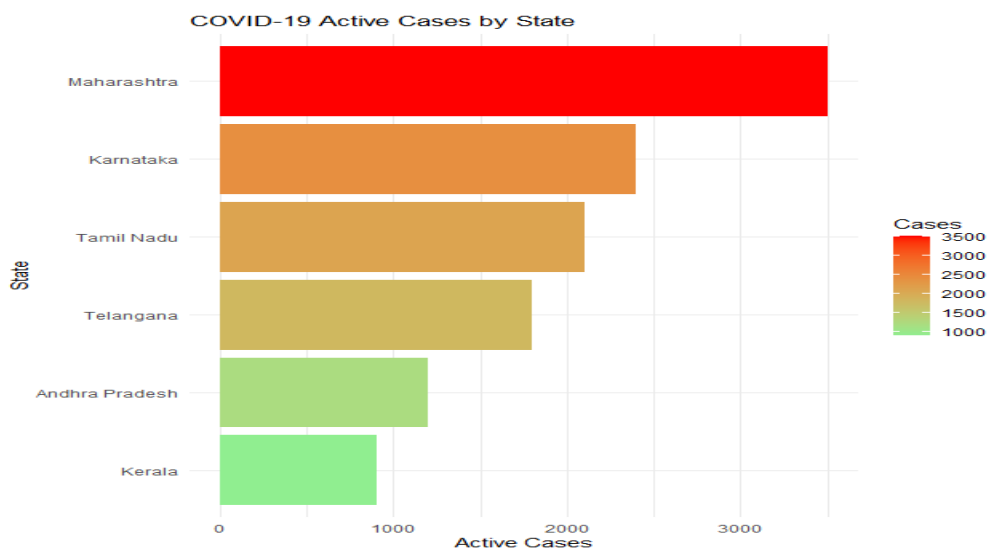
COVID-19 State-wise Analysis

```
library(ggplot2)

covid <- data.frame(
  State = c("Andhra Pradesh", "Telangana", "Karnataka", "Tamil Nadu",
"Kerala", "Maharashtra"),
  ActiveCases = c(1200, 1800, 2400, 2100, 900, 3500)
)

ggplot(covid, aes(x = reorder(State, ActiveCases),
  y = ActiveCases,
  fill = ActiveCases)) +
  geom_col() +
  coord_flip() +
  scale_fill_gradient(low = "lightgreen", high = "red") +
  labs(
    title = "COVID-19 Active Cases by State",
    x = "State",
    y = "Active Cases",
    fill = "Cases"
  ) +
  theme_minimal()
```

OUTPUT:



Scenario 10

Title

Iris Flower Classification

CODE:

```
library(ggplot2)

ggplot(iris,
       aes(x = Sepal.Length,
           y = Petal.Length,
           color = Species,
           shape = Species,
           size = Sepal.Width)) +
  geom_point(alpha = 0.8) +
  labs(
    title = "Iris Flower Classification",
    subtitle = "Species Separation using Multiple Aesthetics",
    x = "Sepal Length (cm)",
    y = "Petal Length (cm)",
    size = "Sepal Width"
  ) +
  theme_classic()
```

OUTPUT:

